

Previous Project Changes/Outline:

I am expanding on my 2nd project again, "The Horde" where I am expanding the main starting level as well based on peer review feedback while also including a player following camera and adding one final weapon pickup, the minigun; it is a weapon drop with a long pickup window of 40 seconds and differs from the SMG pickup by being able to pass through multiple enemies and even walls! I am also tackling the save/load bug within the game, soft-locking.

Introduction:

This is a top-down 2D shoot 'em up-style game, where the player can utilize omnidirectional movement with simple WASD controls. The player can also shoot in the direction of the mouse cursor and fire with the left click. The player can pick up a shotgun power-up(every 2 waves completed), which enables the weapon to fire in a wide spread in front of the mouse cursor, providing more efficiency against a horde of grouped enemies, or a random spawning SMG power-up that allows a faster rate of fire or if more firepower is required the newly added minigun that is able to shred through a horde of enemies with a single bullet! The player's goal is to eliminate all of the enemies within each wave to progress to the next wave, before reaching the final wave that contains a final boss, there are a total of 6 waves. You are given a set health bar with a max HP of 20, with enemies hitting for 10 hit points, and if you reach zero HP, then it is game over! You can pick up HP power-ups to replenish your health should you get too close to death.

Scripts:

TopDownMovement:

The movement script allows the player to move in any given direction that the mouse is facing while also making the sprite face towards the mouse cursor at the same time.

Shooting:

Allows the player to instantiate a projectile from a gameObject attached to the player gameObject, depending on the location of the user's mouse cursor, enabling them to eliminate any enemies visible on screen, and changes weapon pattern depending on what weapon the player GameObject currently has.

Projectile:

Gives GameObjects created from the player GameObject a set velocity, creating a feeling of a weapon being fired, and avoids any possible accidental destruction if projectiles hit a certain GameObject with dependent tags.

VolumeSettings:

Allows the user to adjust volume-dependent GameObjects such as the background ambience and sound effects created from other GameObjects.

AudioManager:

Creates AudioClips for different GameObjects for the ability to sort different sound effects together, stops certain sounds from playing depending on the scene, and plays sounds in certain scenes.

GameManager:

This script handles all of the game logic, such as the spawning of the shotgun powerup, enemies spawned, how many enemies are increased per wave, conditions that are met for the player to win or lose, and the ability to save and load the game at any instance during play.

UIManager:

This script allows the possibility of the pause menu to freeze the game time whenever input is received, allowing the player to save/load, resume, or simply exit to the main menu, and handle the logic for certain panels to be shown depending on the player meeting certain criteria.

PlayerData:

Stores the location of the player to be loaded after a save and also handles the location of the shotgun powerup to be saved as well, while resetting the player if the player has a shotgun but loads to a previous save before the shotgun.

SaveSystem:

Stores the save file path after saving for later use when the player loads up the previous save.

PlayerHealth:

This script creates a health-based system that allows the player to be able to take an additional attack from an enemy before dying if a health pickup is not consumed, and creates a health bar on the UI to give a visual representation to the player on how much damage they have sustained.

HealthPickup:

This script creates a consumable similar to the weapon pickups, except it has a very small chance to drop from eliminating enemies, and if consumed, restores the player to full health.

Enemy:

This script handles the AI of the enemy GameObjects with added collisions if too close to the player GameObject, resulting in a game over, and also handles their pathing depending on the GameObject between the player and enemy GameObjects.

GameObjects:

GameManager:

This GameObject handles all important aspects of the game, as it allows the creation of enemy prefabs, tracks the number of waves with enemies left in each one, decreases the enemies left after enemy GameObjects are destroyed, with a set timer after all enemies are eliminated, before starting the next wave, an important aspect of the core of the game.

Player:

This is an important GameObject as it is what makes the game even start; the entire game loop revolves around the Player GameObject and how it interacts with the other GameObjects, such as the enemies, audio playing on certain triggers from the Player GameObject, health pickups to restore health, and direct interaction with the GameManager GameObject.

Enemy:

This GameObject is another important GameObject, as it is the main objective of the Player GameObject, with both working as one, as the Player GameObject must destroy all enemy GameObjects to progress onto the next wave, can take away health from the Player GameObject, forcing them to interact with the HealthPickup GameObject, which is another interaction with the GameManager GameObject that the enemy GameObject has.

HealthPickups:

This GameObject ties the health and health pickup scripts into one nice GameObject that adds more complexity to the game, with the player being able to survive an additional hit, allowing them to continue fighting with added tension if dragged down to one final hit, and being able to consume these pickups to bring them back to full health.

PowerUps:

These GameObjects directly interact with the Player GameObject, which allows the player to either pick up a weapon GameObject that can be either a ShotgunPowerUp, SMGPowerUp GameObject, MinigunPowerUp or HealthPickup GameObject. These GameObjects change the course of the game by relieving some of the pressure that the player may face, such as being able to wipe out a horde faster with the SMG, Minigun that is capable of wiping out multiple undead at once or those hiding behind walls, or restore health with the HealthPickup.

Project Requirements:

- Player Input that is explained to the player
 - Player input is explained to the player at the main menu screen
- Saving and Loading used for settings and gameplay
 - Saving and loading can be done while in game at any moment during gameplay via the GameMangaer script.
- Multiple sounds
 - All pickups contain their own sound effect, including sound effects for the player, such as taking damage and shooting.
- Settings (including volume)
 - The settings menu allows the player to adjust the audio of either the music or the sound effects.
- Multiple animated objects
 - Player and health pickup contain working animations for enhanced visual gameplay.
- Main Menu, Gameplay Scene(s), Pausing
 - Ability to close application from main menu
 - The player can close the application via the menu.
 - Unpause
 - Ability to pause and unpause the game via the menu button in-game or with the “ESC” key.
 - Exit to main menu
 - Ability to exit to the main menu in-game via the menu button or with the “ESC” key.
- A more complicated scope of gameplay features than Assignment 2
 - A dynamic moving camera that follows the player with a much larger map with an even larger number of enemies per wave, alongside a new minigun powerup that allows a different approach to gameplay with the ability to penetrate through walls.
- Some form of emergent gameplay or progression is provided
 - This may appear as multiple levels that get completed in succession
 - The player progresses through different waves with a final boss on the last wave.
 - It could also mean having a variety of players/weapons and enemies to populate the game with
 - Multiple enemies per wave with a wide variety of weapons to use as power-ups.
- Win and/or Lose conditions that are explained to the player
 - Win and lose conditions are explained to the player at the start menu of the game upon startup.

Saving and loading is satisfied with the SaveSystem script as well as the playerData and enemyData scripts, as they save the locations of the player, enemy, powerups, and health pickups before interaction to be loaded to their saved state.

The player GameObject is animated to represent being idle/moving as the GameObject moves throughout the level, and the HealthPickup is also animated to give an eye-catching visual representation to the player to showcase that the object is important.

Scenes:

The scenes within the game are the MainMenu, Game, and Settings, with the MainMenu connecting both the Game and Settings scenes into one and allowing the player to start the game, adjust settings via the Settings scene, or exit the game, and the Settings scene allows the player to customize the audio to their liking.

The player's objective is to survive a set number of waves with increasing difficulty using a variety of power-ups and health consumables.

Some game elements that indicate the presence of progression are the increasing amount of enemies per wave that lead up to a final boss fight, in which the player can achieve victory if the boss is defeated or lose if the player dies during any of the waves, with the added powerups of a shotgun and SMG every 2 waves, or the health pickups that drop from enemies defeated to restore health that is missing, indicating emergent gameplay as it changes the rules for a short amount of time encouraging a different style of gameplay, one that may be more aggressive and played from afar or a more run-and-gun playstyle with the SMG powerup when compared to a passive playstyle with a weaker weapon (slow-firing pistol).

Incomplete Features:

Some incomplete features are various levels after each completion, and a new enemy type that is ranged, making the player unable to heal for a few seconds.

References:

“Assets Free Laser Bullets Pack 2020 by Wenrex.” *Itch.io*, wenrex.itch.io/laser2020/download/eyJpZCI6NzU2MjQ0LCJleHBpcmVzIjoxNzU5OTkyMjAwfQ%3d%3d.HCIopflAELC0uq6lDfvjmBtl0VY%3d. Accessed 9 Oct. 2025. - Provided bullet asset

YouTube, YouTube, 13 Oct. 2022, www.youtube.com/watch?v=POcQy8aZ6Uw&list=PLx7AKmQhxJFajrXez-0GJgDIKELabQQHT&index=4. - aided in TopDownMovement script

YouTube, YouTube, 7 Mar. 2023, www.youtube.com/watch?v=G-JUp8AMEx0. - Provided scripts and references for audio settings/menu UI, as-is

“Start Menu - 2D Platformer Unity #28.” *YouTube*, YouTube, 12 Mar. 2024, www.youtube.com/watch?v=paaBTt5GcMU. - Menu UI Script, modified for settings menu

“How to Add MUSIC and SOUND EFFECTS to a Game in Unity | Unity 2D Platformer Tutorial #16.” YouTube, YouTube, 25 Feb. 2023, www.youtube.com/watch?v=N8whM1GjH4w. - added the ability to introduce sound

"080902 Shotgun." *Pixabay*, pixabay.com/sound-effects/080902-shotgun-39753/.

"Pistol Shot." *Pixabay*, pixabay.com/sound-effects/pistol-shot-233473/.

"Horror Ambience." *Pixabay*, pixabay.com/music/search/horror%20ambience/. - audio sound effects

“5 Minute Top Down Shooter Unity Tutorial.” *YouTube*, YouTube, 31 Mar. 2022, www.youtube.com/watch?v=LqrAbEaDQzc. - provided shooting script, as-is

"Realistic Shotgun Cocking Sound." *Pixabay*, pixabay.com/sound-effects/realistic-shotgun-cocking-sound-38640/. - shotgun cock sound effect

Pucci Games. "Stones Brick Textures." *itch.io*, pucci-games.itch.io/stones-brick-textures. - provided floor and brick textures

Tokka. "Top-Down Basic Set." *itch.io*, tokka.itch.io/top-down-basic-set/download/eyJpZCI6MTc1MDEsImV4cGlyZXMiOjE3NjAxNDIyNjJ9.cKAOWqK2WUsWDCkdH%2fmfy3tEpKU%3d. - zombie asset

“Unity Player Health and Health Bar Tutorial.” YouTube, www.youtube.com/watch?v=bRcMVkJS3XQ. - PlayerHealth script

gamedev:hq. “Beginners Guide to a Simple Health Pick up System in Unity.” YouTube, 17 July 2025, www.youtube.com/watch?v=pgvj6fcurbw. Accessed 27 Oct. 2025. - HealthPickup

“Heart Pickup Animated.” *Itch.io*, 2024, studionamepending.itch.io/heart-pickup-animated/download/eyJleHBpcmVzIjoxNzYxNzEyNDYzLCJpZCI6MjA5OTc2MH0%3d.gBJqujBp2vMcJL8VRK2KdkH44IU%3d. Accessed 27 Oct. 2025. - Health Pickup sprite/animation

Pixabay. "Health Pickup | Royalty-Free Music." Pixabay.com, 14 Mar. 2023, pixabay.com/sound-effects/health-pickup-6860/. - Health Pickup audio effect

"Pixabay." Pixabay.com, 2022, pixabay.com/sound-effects/zombie-packclean-record-70768/. Accessed 28 Oct. 2025. - Enemy death sound effect

"Pixabay." Pixabay.com, 2022, pixabay.com/sound-effects/pain-noises-98469/. Accessed 28 Oct. 2025. - Pain sound effect

"Pixel Weapons Pack." Itch.io, 2025, asgaard42.itch.io/pixel-weapons-pack/download/eyJleHBpcmVzIjoxNzYxNzE3NjMyLCJpZCI6OTM2ODc0fQ%3d%3d.cxvjIH0Vja5d2WeNQ5kEBgJLsy4%3d. Accessed 28 Oct. 2025. - SMG asset

"Pixabay." Pixabay.com, 2024, pixabay.com/sound-effects/gunshot-smg-shot-1-203471/. Accessed 28 Oct. 2025. - SMG audio

"Securly - Geolocation Sharing." Pixabay.com, 2025, pixabay.com/sound-effects/smg-magazine-out-43792/. Accessed 28 Oct. 2025. - SMG audio

<https://discussions.unity.com/t/destroy-objects-after-a-set-time/230226> - aided in Minigun projectile script

<https://vladpenn.itch.io/weapon> and <https://pixabay.com/sound-effects/search/minigun/> - Minigun sprite and sound effect

<https://www.youtube.com/watch?v=kV9rVinFyAk> - Camera Script